

Routing Errors Matter: An Empirical Analysis of Conditional Hierarchical Dermoscopic Classification

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Abstract—Conditional hierarchical classifiers can reflect broad-to-specific diagnostic organisation, but an incorrect early decision may prevent the appropriate downstream classifier from receiving an image. This work evaluates that routing dependency using a two-stage EfficientNet-B0 hierarchy and two flat comparators on the same leakage-aware, locked 3,668-image internal-test cohort derived from ISIC 2019. The endpoint classes are non-malignant, melanoma, basal cell carcinoma, and squamous cell carcinoma.

Flat DenseNet-121 produced the highest descriptive point estimates for accuracy (0.7914), macro-F1 (0.6351), and weighted-F1 (0.7862), whereas flat EfficientNet-B0 achieved the highest balanced accuracy (0.6503). The predicted-gate hierarchy obtained 0.7402 accuracy and 0.6054 macro-F1. Paired analyses supported DenseNet-121’s accuracy and weighted-F1 gains over both EfficientNet-B0 systems, while its macro-F1 advantage was supported only against the hierarchy. Its balanced accuracy was lower than flat EfficientNet-B0, while the difference from the hierarchy was inconclusive. In the backbone-matched EfficientNet-B0 comparison, the flat-minus-hierarchy macro-F1 difference was 0.0139 with a paired-bootstrap 95% confidence interval of $[-0.0143, 0.0420]$. Exact McNemar testing yielded $p = 0.8207$, detecting no statistically significant difference.

Replacing the learned route with the ground-truth route increased hierarchical macro-F1 to 0.7937, corresponding to a routing-associated loss of 0.1883. Stage 1 blocked 20.079% of malignant images before subtype classification. DenseNet-121 nevertheless achieved only 0.2979 SCC recall. Similar aggregate results can therefore conceal substantial routing and minority-class failure.

Index Terms—dermoscopy, skin lesion classification, hierarchical classification, conditional inference, routing errors, EfficientNet, DenseNet, class imbalance

I. INTRODUCTION

Skin-lesion classes can exhibit overlapping visual characteristics in dermoscopic images, making automated multiclass classification difficult. Deep convolutional neural networks have demonstrated strong representation learning for skin-lesion analysis [1], while International Skin Imaging Collaboration (ISIC) challenges have supported standardised development and evaluation on public dermoscopic collections [2].

Class imbalance remains a central problem. A classifier may achieve high overall accuracy by recognising the majority class while performing poorly on clinically important but under-represented categories. Evaluation should therefore include balanced accuracy, macro-F1, and class-specific recall rather than relying on accuracy alone.

Most multiclass systems make a single flat decision over the complete label set. A conditional hierarchy instead divides the problem into broad and fine-grained decisions. In the present setting, Stage 1 distinguishes non-malignant from malignant lesions, and Stage 2 classifies predicted malignant lesions

as melanoma, basal cell carcinoma (BCC), or squamous cell carcinoma (SCC).

This organisation introduces a routing dependency. A malignant lesion incorrectly predicted as non-malignant is blocked before subtype classification, while a non-malignant lesion incorrectly forwarded to Stage 2 must receive a malignant label. Consequently, an apparently small difference between flat and hierarchical endpoint scores may conceal a substantial loss caused by the gate.

This paper studies that failure mode on a locked internal-test cohort derived from ISIC 2019. The backbone-matched task-decomposition comparison uses EfficientNet-B0 [3] for both a direct four-class model and a two-stage hierarchy. The comparison does not isolate task decomposition alone, because the selected Stage 2 classifier used class-balanced focal loss, whereas the flat EfficientNet-B0 model used ordinary cross-entropy. A protocol-matched flat DenseNet-121 [4] is additionally evaluated on the same split and test cohort. Only the convolutional backbone changes in this comparator; preprocessing, cross-entropy training protocol, optimiser, scheduler, validation-based checkpoint selection, and locked-test policy are preserved.

All three systems were evaluated on the same locked test cohort. Paired bootstrap confidence intervals were used to quantify uncertainty in metric differences for each pairwise comparison, while exact McNemar tests assessed differences in paired correctness. The results are interpreted separately for each metric rather than as evidence of uniform model superiority.

To examine the internal hierarchical failure mechanism, an oracle-routing analysis retains the trained Stage 2 classifier but replaces the predicted Stage 1 route with the ground-truth broad category. The resulting difference estimates routing-associated loss while holding the downstream classifier fixed. Route-specific rates further quantify malignant images blocked at Stage 1, non-malignant images forwarded to Stage 2, and subtype errors among correctly routed malignant images. Oracle routing is diagnostic only and is not a deployable strategy.

A separate experiment evaluates Tis, T1, T2, T3, and T4 prediction on an ISIC-derived melanoma T-category subset. It was not integrated into the primary hierarchy and is reported only as standalone feasibility evidence under severe label scarcity.

The contributions are:

- 1) a same-cohort comparison of flat EfficientNet-B0, flat DenseNet-121, and a conditional EfficientNet-B0 hierarchy;

- 2) paired statistical comparisons among the three systems using ground-truth-class-stratified bootstrap confidence intervals and exact McNemar tests; and
- 3) an oracle-routing and route-specific decomposition that quantifies the contribution of Stage 1 gating errors to hierarchical endpoint performance.

The remainder of this paper reviews related work, formalises the prediction tasks, describes the experimental protocol, reports the results and limitations, and concludes with priorities for future validation.

II. RELATED WORK

A. Dermoscopic Classification and Backbones

Public ISIC resources and HAM10000 have supported reproducible research on dermoscopic classification [2], [5]. EfficientNet jointly scales network depth, width, and resolution and provides compact baseline architectures [3]. DenseNet connects each layer to preceding feature maps, encouraging feature reuse and information flow [4]. In this work, EfficientNet-B0 provides the backbone-matched flat-versus-hierarchy comparison, while DenseNet-121 tests whether the metric-specific findings persist under a protocol-matched alternative backbone.

B. Class Imbalance

Focal loss reduces the influence of well-classified examples and emphasises difficult observations [6]. Class-balanced loss derives weights from the effective number of samples and is designed for long-tailed data [7]. Such methods remain insufficiently characterised by accuracy alone; macro-F1, balanced accuracy, confusion matrices, and class-specific recall are necessary to expose minority-class behaviour.

C. Hierarchical Classification

Top-down hierarchical classifiers are vulnerable to error propagation because an incorrect decision at an upper level constrains all subsequent predictions along the selected path [8]. Hsu and Tseng incorporated diagnostic hierarchy information through hierarchy-aware contrastive learning [9]. More recently, Hsu and Tseng proposed LightDPH, a lightweight hierarchical contrastive-learning framework that jointly considers broad lesion groups and fine-grained classes without relying on a conditional inference gate [10]. Suleiman *et al.* evaluated a two-step hierarchical skin-lesion system combining transfer learning and conventional classifiers [11]. These approaches demonstrate the potential of diagnostic structure, but representation-level hierarchy and conditional routing are distinct. In a conditional system, an upstream decision determines whether a downstream classifier is executed, creating an explicit error-propagation path.

The present study therefore evaluates both endpoint performance and the routing mechanism. Predicted routing is compared with oracle routing while the trained downstream classifier remains fixed, allowing routing-associated loss to be separated from subtype errors after correct routing.

D. Paired Statistical Evaluation

Predictions from models evaluated on the same images are paired. Paired bootstrap resampling can estimate uncertainty in metric differences [12], while McNemar's test evaluates disagreement in paired correctness outcomes [13]. These procedures were applied to all pairwise model comparisons on the common locked test cohort, with metric-specific interpretation of the resulting confidence intervals and paired correctness tests.

III. PROBLEM FORMULATION

Let

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, \quad y_i \in \mathcal{Y} = \{\text{NM}, \text{MEL}, \text{BCC}, \text{SCC}\}, \quad (1)$$

where NM denotes the consolidated non-malignant class.

A. Flat and Hierarchical Predictions

A flat classifier maps every image directly to one endpoint class:

$$\hat{y}_i^{\text{flat}} = \arg \max_{c \in \mathcal{Y}} p(c | x_i). \quad (2)$$

Two flat implementations are evaluated: $f_{\text{flat},E}$ using EfficientNet-B0 and $f_{\text{flat},D}$ using DenseNet-121.

The hierarchy contains a routing classifier

$$f_1 : \mathcal{X} \rightarrow \{\text{NM}, \text{M}\}, \quad (3)$$

and a malignant-subtype classifier

$$f_2 : \mathcal{X} \rightarrow \{\text{MEL}, \text{BCC}, \text{SCC}\}. \quad (4)$$

Its endpoint prediction is

$$\hat{y}_i^{\text{hier}} = \begin{cases} \text{NM}, & f_1(x_i) = \text{NM}, \\ f_2(x_i), & f_1(x_i) = \text{M}. \end{cases} \quad (5)$$

The three systems are evaluated on the same locked test cohort. Formal paired analyses are conducted for all pairwise model comparisons.

B. Oracle Routing

Oracle routing preserves the trained Stage 2 model but uses the ground-truth broad category:

$$\hat{y}_i^{\text{oracle}} = \begin{cases} \text{NM}, & y_i = \text{NM}, \\ f_2(x_i), & y_i \in \{\text{MEL}, \text{BCC}, \text{SCC}\}. \end{cases} \quad (6)$$

For metric M , routing-associated loss is

$$L_{\text{route}}^{(M)} = M(\hat{y}^{\text{oracle}}) - M(\hat{y}^{\text{hier}}). \quad (7)$$

The primary decomposition uses macro-F1. Additional rates measure: (i) malignant images predicted as non-malignant at Stage 1; (ii) non-malignant images forwarded to Stage 2; and (iii) incorrect subtypes among correctly routed malignant images. Oracle routing is a diagnostic upper bound and does not represent an implementable inference policy.

TABLE I
PRIMARY FOUR-CLASS DATASET DISTRIBUTION.

Class	Train	Validation	Test
Non-malignant	11,193	2,398	2,398
Melanoma	3,164	678	678
BCC	2,327	498	498
SCC	440	94	94
Total	17,124	3,668	3,668

C. Standalone T-Category Task

The secondary task uses melanoma images with labels

$$t_j \in \{Tis, T1, T2, T3, T4\}. \quad (8)$$

A separate classifier f_T predicts these labels. It is evaluated independently and is not defined as an integrated third stage of the primary hierarchy.

IV. METHODOLOGY

A. Primary Dataset and Split

The primary experiments used the 25,331-image ISIC 2019 training release. The aggregate collection includes images originating from HAM10000, BCN20000, and MSK sources [5], [14], [15].

The original labels were mapped as follows: NV, BKL, DF, and VASC formed the non-malignant class; MEL mapped to melanoma; BCC mapped to BCC; and SCC mapped to SCC. Actinic keratosis was excluded rather than forced into a malignant or non-malignant endpoint. Four images belonging to label-conflicting duplicate components were also excluded, leaving 24,460 eligible four-class images.

Images connected by a shared non-empty lesion identifier or identical SHA-256 image hash were joined into split groups and assigned intact to the training, validation, or internal-test partition using a diagnosis-stratified 70/15/15 procedure with seed 42. The audit found no cross-partition overlap in split-group identifiers, available lesion identifiers, or exact hashes. However, patient identifiers were unavailable and 2,084 images lacked lesion identifiers. The procedure is therefore described as leakage-aware or known-relation-disjoint, not fully patient-independent.

All principal models were evaluated on the same locked 3,668-image test partition.

B. Preprocessing

All models used 224×224 RGB inputs and ImageNet normalisation. Training transformations consisted of random resized cropping with scale 0.85–1.00 and aspect ratio 0.90–1.10, horizontal and vertical flips with probability 0.5, rotation within $\pm 15^\circ$, and mild colour jitter. Validation and test preprocessing deterministically resized the shorter side to 256 pixels and applied a 224×224 centre crop. No stochastic validation or test augmentation was used.

C. Models and Conditional Inference

Figure 1 illustrates the backbone-matched EfficientNet-B0 comparison.

The flat EfficientNet-B0 model used an ImageNet-initialised backbone with a four-output classifier and ordinary cross-entropy. The flat DenseNet-121 model used the same split, transformations, cross-entropy protocol, optimiser, scheduler, validation-selection metric, seed, and locked-test policy; only the backbone implementation changed.

The hierarchy used two independently trained ImageNet-initialised EfficientNet-B0 models. Stage 1 had two outputs and used ordinary cross-entropy. Stage 2 had three outputs ordered as melanoma, BCC, and SCC. Cross-entropy, inverse-frequency weighted cross-entropy, and class-balanced focal loss were compared using validation macro-F1. The selected Stage 2 model used class-balanced focal loss with effective-number parameter $\beta = 0.9999$ and focal parameter $\gamma = 2.0$:

$$\mathcal{L}_{CBF} = -w_y(1 - p_y)^\gamma \log p_y. \quad (9)$$

At inference, Stage 1 ran for every image. An argmax non-malignant prediction terminated the path, while an argmax malignant prediction invoked Stage 2. No internal-test threshold tuning was performed.

D. Training Protocol

Full runs used batch size 64, at most 30 epochs, and automatic mixed precision on an NVIDIA Tesla T4. AdamW used learning rate 3×10^{-4} and weight decay 10^{-4} . Cosine annealing reduced the learning rate toward 10^{-6} . All experiments used seed 42, early-stopping patience seven, and validation macro-F1 for checkpoint selection. The internal test was not used for loss, threshold, hyperparameter, or checkpoint selection.

The frozen checkpoints corresponded to epoch 2 for flat EfficientNet-B0, epoch 4 for flat DenseNet-121, epoch 5 for Stage 1, and epoch 8 for the selected Stage 2 model.

E. Standalone T-Category Experiment

A candidate identifier list was used only to locate public ISIC records. No images from that external repository or the Dermoscopy Atlas were used. Official ISIC Archive metadata supplied authoritative labels, licensing, and attribution.

Of 856 candidates, 848 were eligible. Broad labels were derived from official diagnosis and Breslow-thickness metadata: in situ as Tis, 0–1 mm as T1, > 1–2 mm as T2, > 2–4 mm as T3, and > 4 mm as T4. The leakage-aware split contained 594 training, 127 validation, and 127 test images. Training counts were 355, 184, 33, 10, and 12 for Tis through T4. The selected EfficientNet-B0 model used inverse-frequency weighted cross-entropy and the epoch-12 validation-selected checkpoint.

F. Evaluation and Statistical Protocol

Accuracy, balanced accuracy, macro-F1, weighted-F1, per-class metrics, and confusion matrices were reported. For all pairwise model comparisons, predictions were aligned by

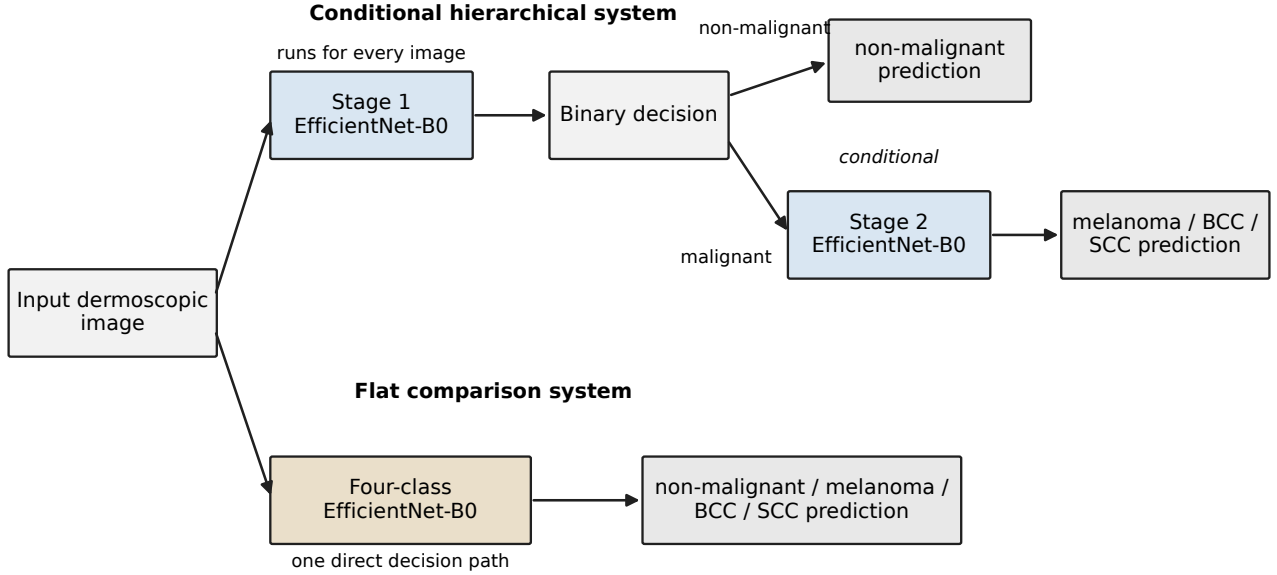


Fig. 1. Backbone-matched EfficientNet-B0 comparison. The flat system directly predicts one of four classes. The hierarchy first predicts non-malignant versus malignant and invokes Stage 2 only for predicted malignant images. The DenseNet-121 flat comparator and standalone T-category experiment are not shown.

TABLE II

FOUR-CLASS INTERNAL-TEST PERFORMANCE. BOLD INDICATES THE HIGHEST POINT ESTIMATE IN EACH COLUMN.

System	Acc.	BAcc.	M-F1	W-F1
DenseNet-121	0.7914	0.6168	0.6351	0.7862
EfficientNet-B0	0.7421	0.6503	0.6192	0.7526
Hierarchy	0.7402	0.6312	0.6054	0.7503

image identifier and analysed using 10,000 ground-truth-class-stratified paired bootstrap replicates with seed 42 and percentile 95% confidence intervals. Exact two-sided McNemar tests were used to assess differences in paired correctness.

V. RESULTS AND DISCUSSION

A. Primary Comparison

Table II reports the three systems on the common locked test cohort.

DenseNet-121 exceeded flat EfficientNet-B0 by 0.0493 accuracy, 0.0159 macro-F1, and 0.0336 weighted-F1, but its balanced accuracy was lower by 0.0335. Relative to the hierarchy, its accuracy and macro-F1 were higher by 0.0513 and 0.0297, respectively, while balanced accuracy was lower by 0.0144.

Thus, the descriptive ranking depends on the evaluation metric. DenseNet-121 had the strongest aggregate point estimates, whereas EfficientNet-B0 had the highest average recall across classes. Paired analyses supported DenseNet-121's accuracy and weighted-F1 gains over both comparators. Its macro-F1 gain was supported only against the hierarchy, while its

TABLE III

CLASS-SPECIFIC PERFORMANCE ON THE LOCKED INTERNAL TEST SET

System	Class	Recall	F1
DenseNet-121	Non-malignant	0.8912	0.8678
	Melanoma	0.5634	0.6025
	BCC	0.7149	0.7265
	SCC	0.2979	0.3436
EfficientNet-B0	Non-malignant	0.7473	0.8218
	Melanoma	0.7537	0.6094
	BCC	0.7811	0.6885
	SCC	0.3191	0.3571
Hierarchy	Non-malignant	0.7794	0.8266
	Melanoma	0.6829	0.5784
	BCC	0.7008	0.6987
	SCC	0.3617	0.3178

balanced accuracy was significantly lower than that of flat EfficientNet-B0.

B. Class-Specific Behaviour

Table III reports recall and F1 for all three systems, while Fig. 2 provides a visual comparison of the backbone-matched flat EfficientNet-B0 model and the conditional hierarchy.

Flat EfficientNet-B0 achieved recalls of 74.7%, 75.4%, 78.1%, and 31.9% for non-malignant, melanoma, BCC, and SCC. The hierarchy increased non-malignant and SCC recall to 77.9% and 36.2%, but melanoma and BCC recall fell to 68.3% and 70.1%. Conditional decomposition therefore did not produce a uniform class-level improvement.

DenseNet-121 achieved recalls of 89.1%, 56.3%, 71.5%, and 29.8% for the same classes. Its high majority-class recall contributed strongly to its aggregate accuracy, while melanoma

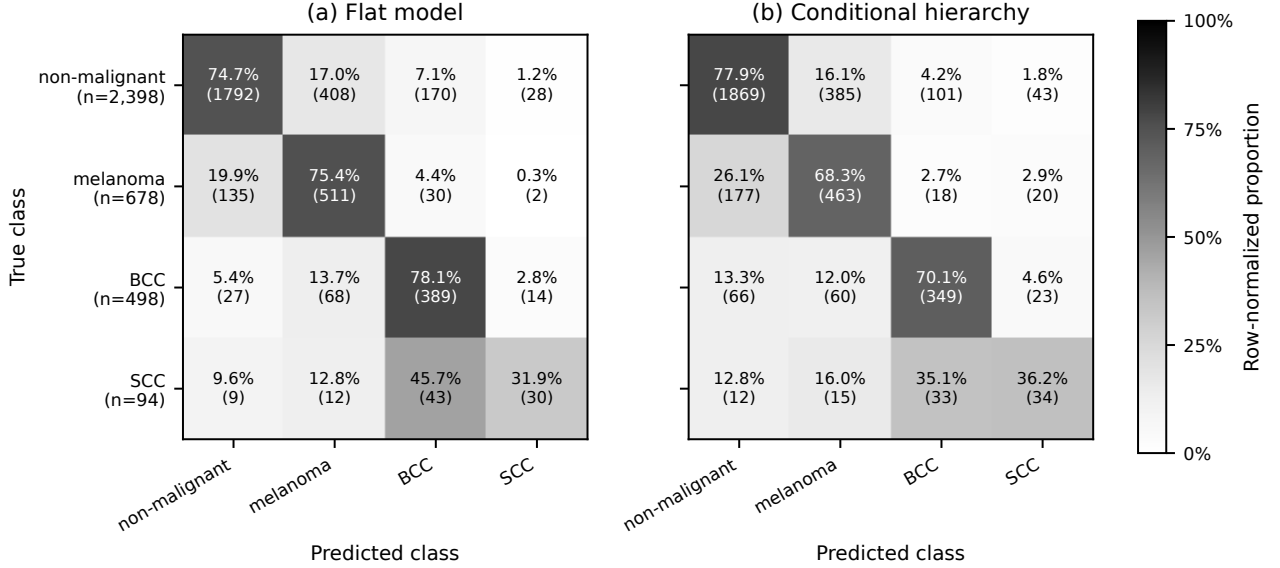


Fig. 2. Row-normalised confusion matrices for flat EfficientNet-B0 and the predicted-gate hierarchy on the common locked test cohort. Counts are shown in parentheses. DenseNet-121 is not included in this figure.

and SCC remained weak. Only 28 of 94 SCC images were correctly classified, giving SCC recall 0.2979 and F1 0.3436. The alternative backbone therefore did not resolve rare-class sensitivity.

C. Paired Model Comparisons

The paired bootstrap estimated the macro-F1 difference (flat minus hierarchy) as 0.013855, with 95% confidence interval $[-0.014255, 0.041963]$. The corresponding accuracy difference was 0.001908 with interval $[-0.011996, 0.015812]$.

The paired correctness table contained 354 images correctly classified only by the flat model and 347 correctly classified only by the hierarchy. Exact two-sided McNemar testing yielded $p = 0.820742$. The analysis therefore did not detect a statistically significant difference between the EfficientNet-B0 systems. This does not establish equivalence, non-inferiority, or identical performance.

Additional paired analyses compared DenseNet-121 with both EfficientNet-B0 systems. Relative to flat EfficientNet-B0, DenseNet-121 increased accuracy by 0.0493 (95% CI $[0.0352, 0.0632]$) and weighted-F1 by 0.0336 ($[0.0198, 0.0471]$), but reduced balanced accuracy by 0.0335 ($[-0.0636, -0.0037]$). Its macro-F1 difference of 0.0159 was inconclusive ($[-0.0148, 0.0463]$). Relative to the hierarchy, DenseNet-121 increased accuracy by 0.0513 ($[0.0371, 0.0654]$), weighted-F1 by 0.0358 ($[0.0221, 0.0495]$), and macro-F1 by 0.0297 ($[0.0003, 0.0588]$). The balanced-accuracy difference of -0.0144 was inconclusive ($[-0.0449, 0.0169]$). Exact McNemar tests also detected differences in paired correctness against flat EfficientNet-B0 ($p = 7.01 \times 10^{-11}$) and the hierarchy ($p = 8.15 \times 10^{-12}$).

TABLE IV
CONDITIONAL-HIERARCHY ROUTING ANALYSIS.

Quantity	Result
Predicted-gate macro-F1	0.605367
Oracle-routing macro-F1	0.793656
Routing-associated loss	0.188289
Malignant blocked at Stage 1	255/1,270 (20.079%)
Non-malignant sent to Stage 2	529/2,398 (22.060%)
Wrong subtype after correct route	169/1,015 (16.650%)

D. Routing-Error Decomposition

Oracle routing increased macro-F1 from 0.6054 to 0.7937. Among 1,270 malignant images, 255 were blocked before subtype classification. Conversely, 529 non-malignant images were sent to a classifier restricted to malignant outputs. These counts correspond to a malignant-class sensitivity of 79.921% and a non-malignant specificity of 77.940% at the Stage 1 gate. Among the 1,015 malignant images correctly routed to Stage 2, 169 received the wrong subtype.

Both Stage 1 routing-error rates exceeded the subtype-error rate among correctly routed malignant images. Together with the 0.1883 macro-F1 gap, these findings indicate that Stage 1 routing errors were a substantial contributor to endpoint performance loss in the implemented hierarchy. The oracle score remains a diagnostic upper bound: it uses ground-truth routing and does not predict what a practically improved gate would achieve.

E. Standalone T-Category Feasibility

The validation-selected weighted-cross-entropy model obtained test macro-F1 0.275611 and balanced accuracy 0.386039 on the 127-image T-category test partition. T2 and T4 recalls

were both zero, with supports of seven and one. The aggregate scores therefore conceal complete failure on the scarcest categories.

This experiment provides standalone feasibility evidence under severe label scarcity. It does not constitute clinically useful staging, deployable melanoma assessment, or an integrated three-stage result.

F. Limitations

The study used one random seed and one internal ISIC-derived cohort, limiting the generalisability of the observed differences. The split controlled available lesion identifiers and exact duplicates, but patient-independent separation could not be guaranteed because patient identifiers were unavailable and 2,084 images lacked lesion identifiers. SCC had only 94 test images, and the standalone T-category task had still smaller rare classes.

No external evaluation, calibration study, explainability experiment, or skin-tone fairness analysis was completed. Direct numerical comparison with studies using different classes, datasets, or splitting procedures would therefore be misleading. The evidence supports an internal routing-error analysis, not clinical validation, external generalisation, or deployment readiness.

VI. CONCLUSION AND FUTURE WORK

This study compared flat EfficientNet-B0, flat DenseNet-121, and a conditional EfficientNet-B0 hierarchy on the same locked 3,668-image internal-test cohort. DenseNet-121 achieved the highest accuracy, macro-F1, and weighted-F1 point estimates, while EfficientNet-B0 retained the highest balanced accuracy. Paired analyses supported DenseNet-121's accuracy and weighted-F1 gains over both comparators, although its performance advantage was not uniform across metrics.

The hierarchy did not improve upon flat EfficientNet-B0 in the reported endpoint metrics, and the paired analysis detected no statistically significant difference. More importantly, oracle routing increased hierarchical macro-F1 from 0.6054 to 0.7937. Stage 1 blocked 20.079% of malignant images and incorrectly forwarded 22.060% of non-malignant images, indicating that routing errors were a substantial source of endpoint performance loss.

Neither the alternative backbone nor the hierarchy resolved rare-class sensitivity; DenseNet-121 correctly classified only 28 of 94 SCC images.

Future work should prioritise repeated-seed evaluation, independently validated routing improvements, larger minority-class cohorts, and external testing on independently collected data. Until then, the systems should be interpreted as internally evaluated experimental models rather than clinically validated or deployment-ready tools.

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